

An intermediate eigenvalue problem in electronic structure calculation

Shao-Liang Zhang[†]

张绍良

[†]Nagoya University

名古屋大学应用物理系

Symmetric Definite

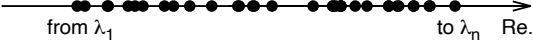
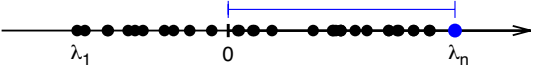
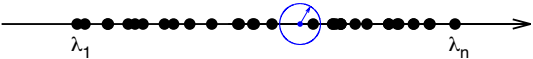
Generalized Eigenvalue Problems

$$Ax_i = \lambda_i Bx_i, \quad x_i \neq 0.$$

IN: $A^T = A, B^T = B \in \mathbb{R}^{n \times n}.$
 $B > 0$ (positive definite).

OUT: $\lambda_i \in \mathbb{R}.$
 $(\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n)$

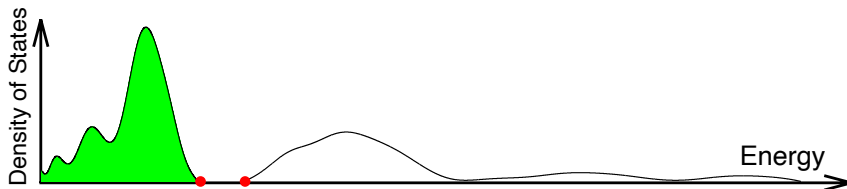
Existing Needs for λ_i

Needs		Methods
All		QZ
Largest Abs.		Lanczos
Near Point		Shift & Invert
Within Domain		Contour Integral

Specific Need: Motivation

Application: Electronic Structure Calculation

Purpose: Theoretical Analysis of Luminescent Materials



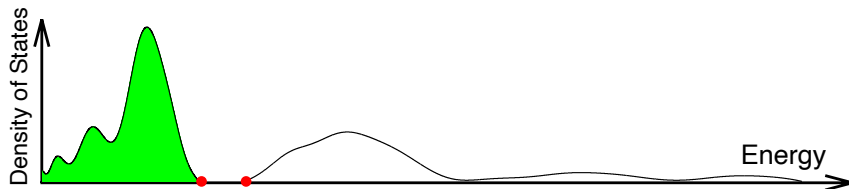
Compute: Certain “Energy Levels” related to luminescence.

From the atomic configuration of the material,
it is determined that which levels need to be computed.

Specific Need: Aim

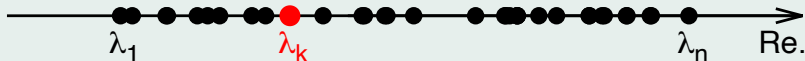
Application: Electronic Structure Calculation

Compute: Certain Energy Levels



Problem: Symmetric Definite GEP $Ax_i = \lambda_i Bx_i$

Target: λ_k with k determined by physical properties



类似例子（聚会抽奖）

参加人数上限为 N （事先准备 N 个写有不同数字的纸团）

实际参加人数为 n ($n < N$, 每人进会场时抽取1个纸团)

抽奖时主持人随意宣布一个整数 k ($1 \leq k \leq n$)

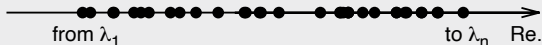
持有 第 k 小 数字（纸团）的人获奖

怎样快速找到这个获奖者？

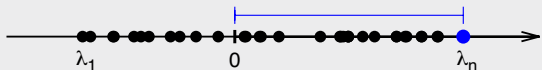
Existing Needs and Methods

Needs

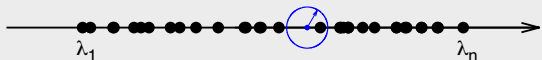
All



Largest
Abs.



Near
Point



Within
Domain



Methods

QZ

Lanczos

Shift &
Invert

Contour
Integral

- 1 **Not** Efficient.
- 2 **Interval** which contains λ_k is necessary.

解决办法之一（2分法，概略）

主持人先说出一个数字 d ，请持有小于 d 数字（纸团）的人举手
（设举手人数为 $n(d)$ ）

如果找到两个数字 a 和 b ($a < b$)，使 $n(a) < k < n(b)$ 成立
（我们就知道 第 k 小 数字的范围）

设 $c = (a+b)/2$ 请持有小于 c 数字（纸团）的人举手
如果 $n(c) < k$ ， $a := c$ ；否则 $b := c$ （范围缩小一半）

重复下去，至到 $n(b) - n(a) = 1$

Our Approach: Outline

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

1 Main Theorem

Sylvester's Law of Inertia

2 Standard EVP

Spectral Bisection

Schematic Example

3 Generalized EVP

Symmetric Positive Definite B

Numerical Example

Our Approach: Outline

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

① **Main Theorem**
Sylvester's Law of Inertia

② **Standard EVP**
Spectral Bisection
Schematic Example

③ **Generalized EVP**
Symmetric Positive Definite B
Numerical Example

Main Theorem

Sylvester's Law of Inertia [Sylvester(1852)]

Let A be a Hermitian matrix of order n .

For any nonsingular matrix $C \in \mathbb{C}^{n \times n}$,
 $B = C^H A C$ and A have the same **inertia**.

Inertia

The inertia of a real symmetrix matrix A is a triple

$$\text{In}(A) = [\pi(A), \nu(A), \zeta(A)],$$

where $\pi(A)$ is the number of positive eigenvalues of A ,
 $\nu(A)$ is the number of negative eigenvalues of A , and
 $\zeta(A)$ is the number of zero eigenvalues of A .

Main Theorem

Sylvester's Law of Inertia

For $A^H = A$ of order n
and any nonsingular
matrix $C \in \mathbb{C}^{n \times n}$,

$$\text{In}(C^H A C) = \text{In}(A).$$

Computing Inertia

LDL^H Decomposition (with pivoting P)

$$PAP^T = LDL^H$$

$$= \begin{bmatrix} 1 & & & & & \\ * & 1 & & & & \\ * & * & 1 & & & \\ * & * & * & 1 & & \\ & & & & * & \\ & & & & & 1 & * & * & * \\ & & & & & & 1 & * & * \\ & & & & & & & 1 & * \\ & & & & & & & & 1 \end{bmatrix}$$

L is a unit lower triangular.

D is a (block) diagonal matrix.

By the inertia theorem,

$$\text{In}(D) = \text{In}(LDL^H)$$

$$= \text{In}(PAP^T) = \text{In}(A).$$

Let D_i denote diagonal elements of D .

$$\text{In}(D) = \left[\sum_i \pi(D_i), \sum_i \nu(D_i), \sum_i \zeta(D_i) \right].$$

Our Approach: Outline

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

① Main Theorem
Sylvester's Law of Inertia

② Standard EVP
Spectral Bisection
Schematic Example

③ Generalized EVP
Symmetric Positive Definite B
Numerical Example

Spectral Bisection

Symmetric

Eigenvalue Problem

$$Ax_i = \lambda_i x_i.$$

IN: $A^T = A \in \mathbb{R}^{n \times n}$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mI$ to LDL^T

$\nu(A - mI) := \nu(D)$

if $\nu(A - mI) < k$ **then**

$a := m, \nu_a := \nu(A - mI)$

else

$b := m, \nu_b := \nu(A - mI)$

end

if $\nu_b - \nu_a = 1$ **then** **break**

end

$\lambda_k := (a + b)/2$

Spectral Bisection

Symmetric Eigenvalue Problem

$$Ax_i = \lambda_i x_i.$$

The initial interval $[a, b]$ contains every eigenvalue of A .

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mI$ to LDL^T

$\nu(A - mI) := \nu(D)$

if $\nu(A - mI) < k$ **then**

$a := m, \nu_a := \nu(A - mI)$

else

$b := m, \nu_b := \nu(A - mI)$

end

if $\nu_b - \nu_a = 1$ **then** break

end

$\lambda_k := (a + b)/2$

Spectral Bisection

Symmetric Eigenvalue Problem

$$Ax_i = \lambda_i x_i.$$

Calculate the midpoint m .

```
Set an initial interval  $[a, b]$ 
 $\nu_a := 0, \nu_b := n$ 
for  $i = 1$  to itrtol do
     $m := (a + b)/2$ 
    Decompose  $A - mI$  to  $LDL^T$ 
     $\nu(A - mI) := \nu(D)$ 
    if  $\nu(A - mI) < k$  then
         $a := m, \nu_a := \nu(A - mI)$ 
    else
         $b := m, \nu_b := \nu(A - mI)$ 
    end
    if  $\nu_b - \nu_a = 1$  then break
end
 $\lambda_k := (a + b)/2$ 
```


Spectral Bisection

Symmetric Eigenvalue Problem

$$Ax_i = \lambda_i x_i.$$

Compute $\mathbf{In}(A - mI)$.
 $P(A - mI)P^T$
 $= LDL^T$

$$= \begin{bmatrix} 1 & & * & & 1 & * & * \\ * & 1 & & * & & 1 & * \\ * & * & 1 & & * & & 1 \end{bmatrix}$$

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

Decompose $A - mI$ to LDL^T

$\nu(A - mI) := \nu(D)$

if $\nu(A - mI) < k$ **then**

$a := m, \nu_a := \nu(A - mI)$

else

$b := m, \nu_b := \nu(A - mI)$

end

if $\nu_b - \nu_a = 1$ **then break**

end

$\lambda_k := (a + b)/2$

Spectral Bisection

Symmetric Eigenvalue Problem

$$Ax_i = \lambda_i x_i.$$

Eigenvalues of $A - mI$ are equal to $\lambda_i - m$.

Hence, $\nu(A - mI)$ is the number of eigenvalues of A smaller than m .

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mI$ to LDL^T

$\nu(A - mI) := \nu(D)$

if $\nu(A - mI) < k$ **then**

$a := m, \nu_a := \nu(A - mI)$

else

$b := m, \nu_b := \nu(A - mI)$

end

if $\nu_b - \nu_a = 1$ **then** **break**

end

$\lambda_k := (a + b)/2$

Spectral Bisection

Symmetric Eigenvalue Problem

$$Ax_i = \lambda_i x_i.$$

$\nu_b - \nu_a$ is the number of eigenvalues within the interval $[a, b)$.

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mI$ to LDL^T

$\nu(A - mI) := \nu(D)$

if $\nu(A - mI) < k$ **then**

$a := m, \nu_a := \nu(A - mI)$

else

$b := m, \nu_b := \nu(A - mI)$

end

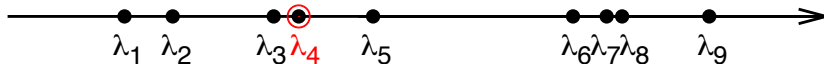
if $\nu_b - \nu_a = 1$ **then** **break**

end

$\lambda_k := (a + b)/2$

Schematic Example

There are 9 simple eigenvalues. λ_4 is needed ($k = 4$).



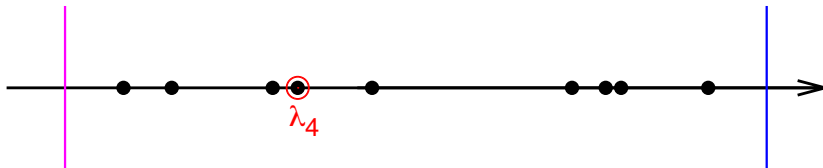
Left End a

Midpoint m

Right End b

Schematic Example

Set an initial interval $[a, b]$.



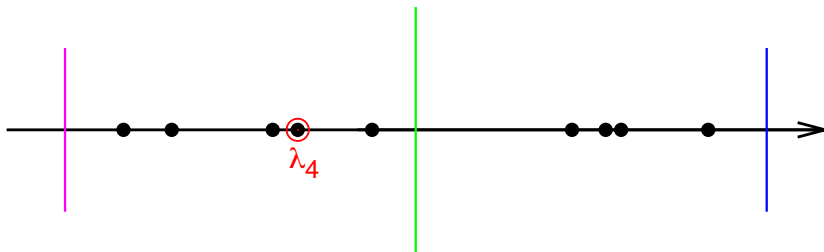
Left End a

Midpoint m

Right End b

Schematic Example

Calculate the midpoint m . Compute $\nu(A - mI)$.



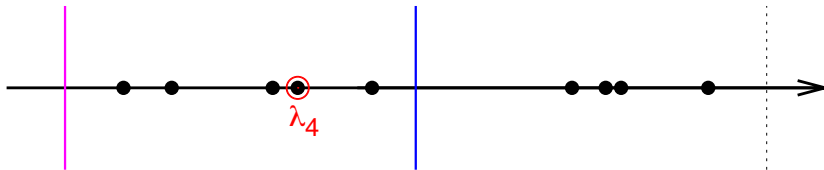
Left End a

Midpoint m

Right End b

Schematic Example

$\nu(A - mI) = 5 \geq k$. Therefore, $b := m$.



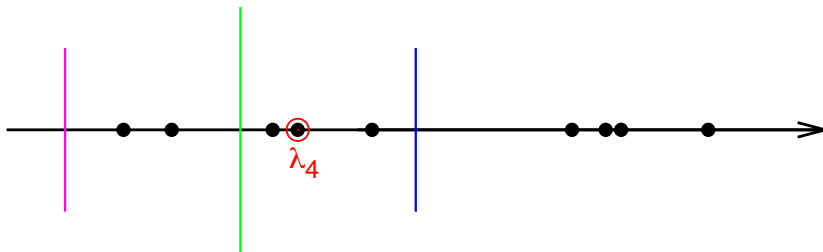
Left End a

Midpoint m

Right End b

Schematic Example

Again, compute the midpoint m and $\nu(A - mI)$.



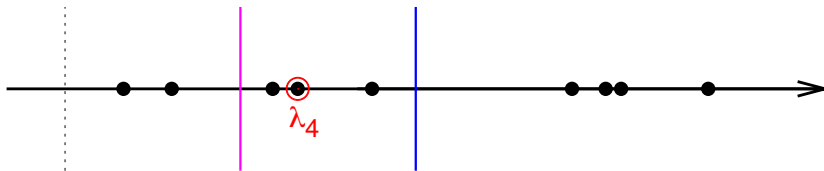
Left End a

Midpoint m

Right End b

Schematic Example

$\nu(A - mI) = 2 < k$. Therefore, $a := m$.



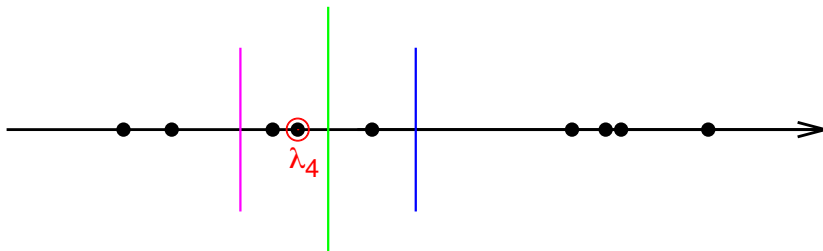
Left End a

Midpoint m

Right End b

Schematic Example

$\nu_b - \nu_a = 3$. Repeat spectral bisection.



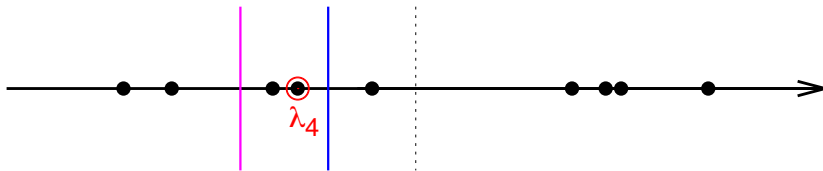
Left End a

Midpoint m

Right End b

Schematic Example

$$\nu_b - \nu_a = 2.$$



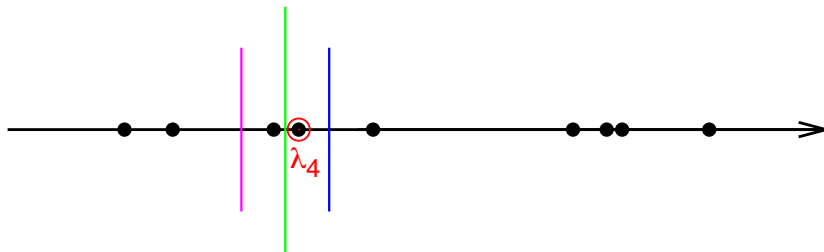
Left End a

Midpoint m

Right End b

Schematic Example

Repeat spectral bisection.



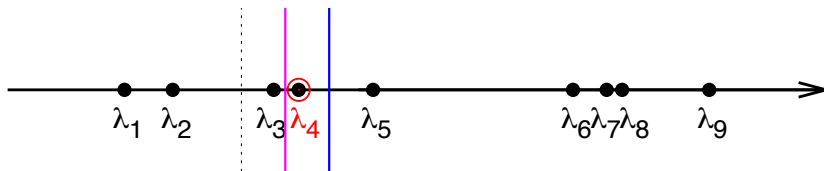
Left End a

Midpoint m

Right End b

Schematic Example

The interval which contains λ_4 only ($\nu_b - \nu_a = 1$).



Left End a

Midpoint m

Right End b

Our Approach: Outline

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

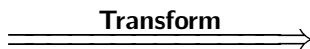
① Main Theorem
Sylvester's Law of Inertia

② Standard EVP
Spectral Bisection
Schematic Example

③ Generalized EVP
Symmetric Positive Definite B
Numerical Example

Extension

Symmetric
Definite
GEP



Symmetric
Eigenvalue
Problem

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.
OUT: $\lambda_k \in \mathbb{R}$.

$$\tilde{A}\tilde{x}_i = \lambda_i\tilde{x}_i.$$

IN: $\tilde{A}^T = \tilde{A}$,
 $k \in \mathbb{N}$.
OUT: $\lambda_k \in \mathbb{R}$.

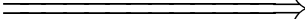
Extension

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A,$
 $B^T = B > 0,$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Transform

Cholesky decomposition
 $B = GG^T.$

Then
 $G^{-1}AG^{-T}G^Tx = \lambda BG^Tx.$

Let
 $\tilde{A} = G^{-1}AG^{-T}, \tilde{x} = G^Tx.$
Note that λ is the same.

Symmetric Eigenvalue Problem

$$\tilde{A}\tilde{x}_i = \lambda_i\tilde{x}_i.$$

IN: $\tilde{A}^T = \tilde{A},$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Extension

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A,$
 $B^T = B > 0,$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

$\xrightarrow{\text{Transform}}$
Cholesky decomposition
 $B = GG^T.$

Then
 $G^{-1}AG^{-T}G^Tx = \lambda BG^Tx.$

Let
 $\tilde{A} = G^{-1}AG^{-T}, \tilde{x} = G^Tx.$
Note that λ is the same.

$\xleftarrow{\text{Bisection}}$

Symmetric Eigenvalue Problem

$$\tilde{A}\tilde{x}_i = \lambda_i\tilde{x}_i.$$

IN: $\tilde{A}^T = \tilde{A},$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Extension

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A,$
 $B^T = B > 0,$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Transform?

Cholesky decomposition

$$B = GG^T.$$

Then

$$G^{-1}AG^{-T}G^Tx = \lambda BG^Tx.$$

Let

$$\tilde{A} = G^{-1}AG^{-T}, \quad \tilde{x} = G^Tx.$$

Note that λ is the same.

Bisection?

Symmetric Eigenvalue Problem

$$\tilde{A}\tilde{x}_i = \lambda_i\tilde{x}_i.$$

IN: $\tilde{A}^T = \tilde{A},$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A,$
 $B^T = B > 0,$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Transform?
 $\xrightarrow{\hspace{1.5cm}}$

By the inertia theorem,
$$\begin{aligned}\operatorname{In}(\tilde{A}) &= \operatorname{In}(G^{-1}AG^{-T}) \\ &= \operatorname{In}(A).\end{aligned}$$

Likewise,

$$\begin{aligned}\operatorname{In}(\tilde{A} - mI) &= \operatorname{In}(G^{-1}(A - mB)G^{-T}) \\ &= \operatorname{In}(A - mB).\end{aligned}$$

$\xleftarrow{\hspace{1.5cm}}$
Bisection?

Symmetric Eigenvalue Problem

$$\tilde{A}\tilde{x}_i = \lambda_i \tilde{x}_i.$$

IN: $\tilde{A}^T = \tilde{A},$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A,$
 $B^T = B > 0,$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Transform?

By the inertia theorem,

$$\begin{aligned}\operatorname{In}(\tilde{A}) &= \operatorname{In}(G^{-1}AG^{-T}) \\ &= \operatorname{In}(A).\end{aligned}$$

Likewise,

$$\begin{aligned}\operatorname{In}(\tilde{A} - mI) &= \operatorname{In}(G^{-1}(A - mB)G^{-T}) \\ &= \operatorname{In}(A - mB).\end{aligned}$$

Bisection!

Symmetric Eigenvalue Problem

$$\tilde{A}\tilde{x}_i = \lambda_i \tilde{x}_i.$$

IN: $\tilde{A}^T = \tilde{A},$
 $k \in \mathbb{N}.$

OUT: $\lambda_k \in \mathbb{R}.$

Spectral Bisection

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mB$ to LDL^T

$\nu(A - mB) := \nu(D)$

if $\nu(A - mB) < k$ **then**

$a := m, \nu_a := \nu(A - mB)$

else

$b := m, \nu_b := \nu(A - mB)$

end

if $\nu_b - \nu_a = 1$ **then** break

end

$\lambda_k := (a + b)/2$

Spectral Bisection

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

The only difference to the standard version is that there is B , in place of I .

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mB$ to LDL^T

$\nu(A - mB) := \nu(D)$

if $\nu(A - mB) < k$ **then**

$a := m, \nu_a := \nu(A - mB)$

else

$b := m, \nu_b := \nu(A - mB)$

end

if $\nu_b - \nu_a = 1$ **then** break

end

$\lambda_k := (a + b)/2$

Spectral Bisection

Symmetric Definite GEP

$$Ax_i = \lambda_i Bx_i.$$

$\tilde{A} - mI$ and $A - mB$
have the same inertia.
Hence, $\nu(A - mB)$ is
the number of eigenvalues
of the matrix pencil
(A, B) smaller than m .

Set an initial interval $[a, b]$

$\nu_a := 0, \nu_b := n$

for $i = 1$ **to** itrtol **do**

$m := (a + b)/2$

 Decompose $A - mB$ to LDL^T

$\nu(A - mB) := \nu(D)$

if $\nu(A - mB) < k$ **then**

$a := m, \nu_a := \nu(A - mB)$

else

$b := m, \nu_b := \nu(A - mB)$

end

if $\nu_b - \nu_a = 1$ **then** break

end

$\lambda_k := (a + b)/2$

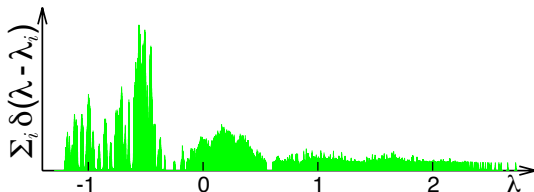
Numerical Example

- $n = 2,048$

Nnz = 2,267,232



- Every eigenvalue is simple.
- Initial interval:
[−3.0, 3.0] as an input
- Stopping Criteria:
 $\nu_b - \nu_a = 1$ & `itrtol` = 50
- LDL^T decomposition:
LAPACK's DSYTRF



Environment

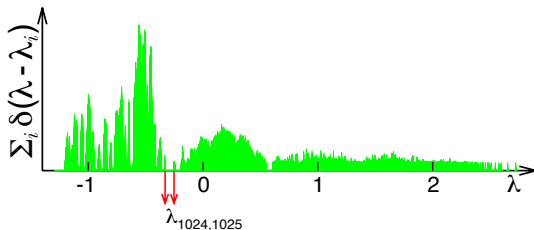
Xeon 2.93GHz (1 core)
with double precision
Compiler: gfortran 4.1.2
Library: ATLAS 3.8.0

Numerical Example

- $n = 2,048$
 $\text{Nnz} = 2,267,232$



- Every eigenvalue is simple.
- Initial interval:
 $[-3.0, 3.0]$ as an input
- Stopping Criteria:
 $\nu_b - \nu_a = 1$ & $\text{itrtol} = 50$
- LDL^T decomposition:
LAPACK's DSYTRF



EV	λ_{1024}	λ_{1025}
Iter.	11	10

0.5 sec / iter.

Environment

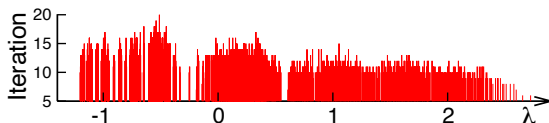
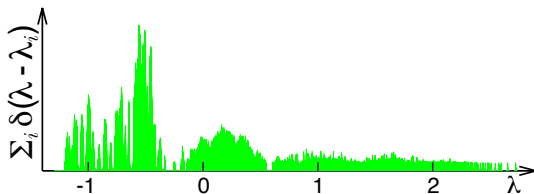
Xeon 2.93GHz (1 core)
with double precision
Compiler: gfortran 4.1.2
Library: ATLAS 3.8.0

Numerical Example

- ▶ $n = 2,048$
 $\text{Nnz} = 2,267,232$



- ▶ Every eigenvalue is simple.
- ▶ Initial interval:
 $[-3.0, 3.0]$ as an input
- ▶ Stopping Criteria:
 $\nu_b - \nu_a = 1$ & $\text{itrtol} = 50$
- ▶ LDL^T decomposition:
LAPACK's DSYTRF



Environment

Xeon 2.93GHz (1 core)
with double precision
Compiler: gfortran 4.1.2
Library: ATLAS 3.8.0

Our Approach: Summary

Symmetric Definite
GEP

$$Ax_i = \lambda_i Bx_i.$$

IN: $A^T = A$,
 $B^T = B > 0$,
 $k \in \mathbb{N}$.

OUT: $\lambda_k \in \mathbb{R}$.

1 Main Theorem

Sylvester's Law of Inertia

2 Standard EVP

Spectral Bisection

Schematic Example

3 Generalized EVP

Symmetric Positive Definite B

Numerical Example

- ▶ **Application-Driven:**
Electronic Structure Calculation.
- ▶ **Specific Need:**
 λ_k with k determined by applications.
- ▶ **Algorithm:**
Bisection with LDL^T Decomposition.
- ▶ **Further Research Needed!**

Thank You